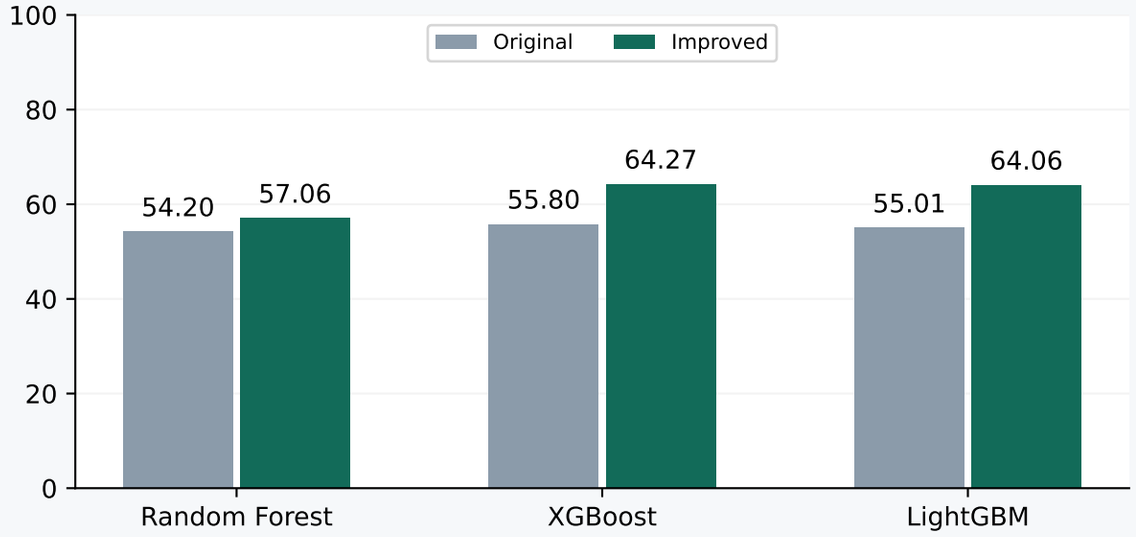
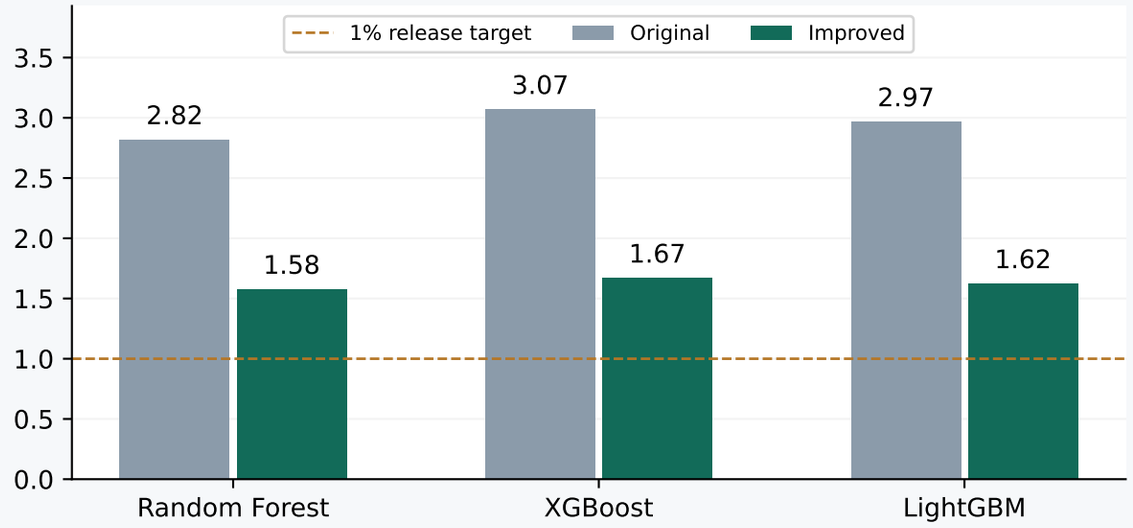
LINKGUARD | Measured model improvements

48,406 test URLs · 33,285 domains unseen by both training corpora · 3 November 2026

Phishing recall (%) — higher is better



False positives (%) — lower is better



Model	Version	Accuracy	Precision	Phishing recall	F1	False positives
Random Forest	Original	82.87%	90.57%	54.20%	67.81%	2.82%
Random Forest	Improved	84.65%	94.76%	57.06%	71.23%	1.58%
XGBoost	Original	83.24%	90.08%	55.80%	68.91%	3.07%
XGBoost	Improved	86.99%	95.05%	64.27%	76.69%	1.67%
LightGBM	Original	83.04%	90.25%	55.01%	68.36%	2.97%
LightGBM	Improved	86.95%	95.17%	64.06%	76.57%	1.62%

Research ensemble: 76.34% phishing recall · 85.08% F1 · 1.55% false positives

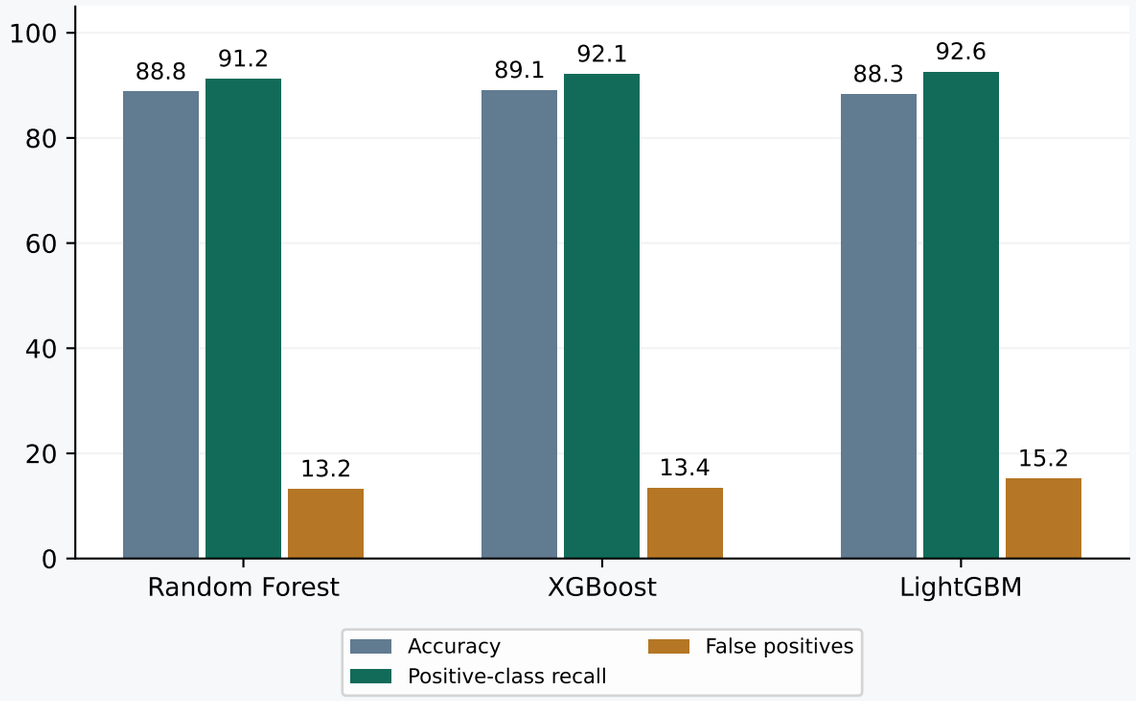
Not deployed: it missed the 1% false-positive gate. The shipped application retains the existing XGBoost model.

Same test cohort and frozen thresholds. Training data/features/policies changed together. No UCI-only shortcut benchmark is used.
Test prevalence is 33.3% phishing. Precision changes with real browsing prevalence; these are benchmark results, not universal guarantees.

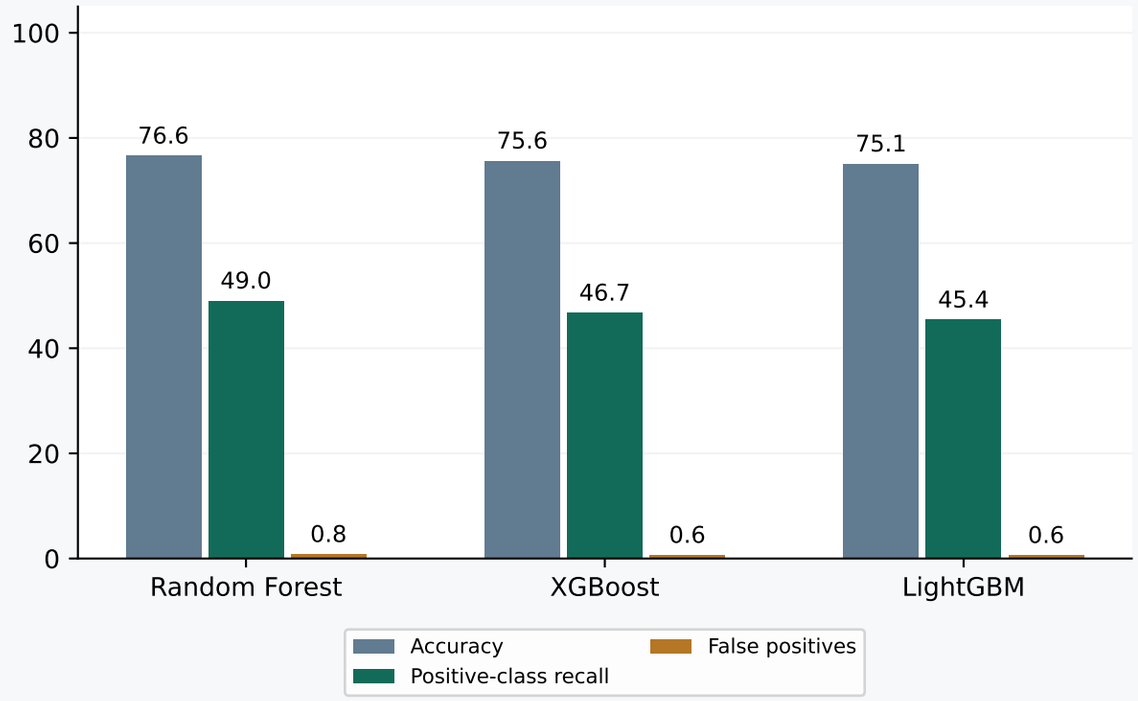
LINKGUARD | Single-source validation

ealvaradob only · 811,472 cleaned URLs · 178,918 test URLs · 71,527 held-out domains

Balanced threshold: maximum validation F1



Conservative: 1% validation FPR target



What the client can conclude

- UCI is excluded from this single-source benchmark; the 32-feature extractor ignores the conventional www prefix.
- Balanced accuracy is 88–89%, with 13–15% false positives. High recall does not mean the system is ready for automatic blocking.
- Conservative thresholds reduce false positives below 1%, but recall falls to 45–49%. Thresholds were chosen on validation only.
- Positive-class labels here follow the publisher; they are not asserted to be independently verified phishing-only ground truth.
- No dataset is proven bias-free. The first page uses separately labelled phishing and domain exclusions for a stronger comparison.
- Original models used legacy mixed-source training. Improved research models used typed data, richer features and separate calibration.
- The CSV contains the exact counts, thresholds, cohort fingerprint and deployment status behind these charts.